

Private Quantum Database

Giancarlo Gatti
Mondragon University
ggatti@mondragon.edu

Floris Geerts
University of Antwerp
floris.geerts@uantwerp.be

Rihan Hai
Delft University of Technology
r.hai@tudelft.nl

Abstract—We ask what a *quantum-native* database can do for data management *today*, and argue that two-sided privacy is a compelling near-term capability. We target single-server symmetric private information retrieval (SPIR), which requires both *user privacy* (hiding which record is queried) and *data privacy* (preventing retrieval beyond the requested record). Our design encodes relational tuples using quantum random access codes (QRACs), producing quantum states that embody the table’s contents. The server sends a bounded number of these states; the client makes one destructive measurement tied to the desired key, reconstructing only that tuple while making unqueried rows physically inaccessible due to measurement disturbance and basis incompatibility. To keep this deployable on current hardware, we propose a hybrid quantum–classical architecture: classical descriptions of the states live in a standard DBMS, and fresh instances are generated at query time. The result is single-server, SPIR-style privacy without trusted hardware or heavyweight cryptography, offering a concrete, application-driven path to introducing quantum components into future database systems.

I. INTRODUCTION

Database systems have been engineered for decades around classical data on classical machines. As quantum computing moves from laboratory curiosity to a commercially funded technology stack, our field faces a foundational question:

What can a quantum database do?

Answering this requires application-driven analysis and realistic system design, while engaging honestly with the limitations and potential of today’s quantum hardware [34]. We are now in the *noisy intermediate-scale quantum* (NISQ) era, characterized by quantum processors with hundreds to low-thousands of qubits that remain noisy and error-prone. As of 2025, representative platforms include 56-qubit trapped-ion processors [31], [78], 127-qubit superconducting processors [64], and neutral-atom processors reaching up to 280 qubits [20]. One of IBM’s most advanced processors is Heron¹, with 156 qubits and no full error correction yet. IBM Quantum System Two, currently under development, connects multiple Heron processors and aims to scale to 4,000 qubits by the end of 2025 and 100,000 qubits in the 2030s. Meanwhile, the two-qubit gate error rates approach 0.1% [31], [47], and single-qubit gate errors are below 0.01% [70], [97]. Keeping pace with the rapid evolution of quantum computing, our goal is to identify the right applications and design a quantum database that delivers meaningful real-world value.

Quantum data is the information represented as quantum states. Such data may be classical data encoded into qubits, e.g., images or vectors embedded for quantum machine learning algorithms [17]; or natively quantum outputs produced by quantum devices such as quantum sensors [55]. Consistent with classical database foundations [2], [35], [39], [94], we define a *quantum database* as an organized collection of quantum data. A *quantum database management system (QDBMS)* is the software that manages quantum data and makes it useful. A *quantum database system* is the end-to-end stack comprising the quantum database, the QDBMS, and the underlying quantum (and supporting classical) hardware. Quantum database systems employ quantum phenomena such as superposition, entanglement, and measurement to realize functionality that is inefficient or impossible for classical database systems. In the NISQ era, practical quantum database systems are most likely to be implemented as hybrid quantum–classical systems [53].

Existing studies indicate three broad motivations of quantum database applications. The first is **database search**. Grover’s algorithm [49] can search unstructured data with complexity of $\mathcal{O}(\sqrt{N})$, where N is the size of the domain. However, recent work indicates that NISQ devices cannot achieve the promised quadratic speedup [26]. Moreover, classical databases have more mature implementations, such as B-trees or hash indexes. Thus, the opportunity window for search acceleration is, for now, narrow. The second possibility is using a quantum database to **store native quantum data**, such as the quantum states emitted directly by quantum sensors. While compelling, these workloads are niche and do not motivate a general-purpose quantum database.

Position. We explore another promising near-term possibility, that is, employ quantum databases for **privacy**. Quantum cryptographic primitives such as quantum key distribution [14], quantum private queries [44], and quantum oblivious transfer [15] have shown potential in regulated domains such as banking and finance [86], [91], privacy-constrained healthcare [60], [105] and biometric verification [95]. Building on these primitives, we investigate employing quantum databases for private information retrieval (PIR) [66] and its stronger variant, symmetric PIR (SPIR) [32], [42]. PIR ensures user privacy: a user queries a record from the database server without revealing to the server which record it is retrieving. SPIR additionally enforces data privacy so the user learns only the requested tuple. PIR is already deployed in production. For example, Apple reports PIR-backed visual search for

¹<https://docs.quantum.ibm.com/guides/processor-types>

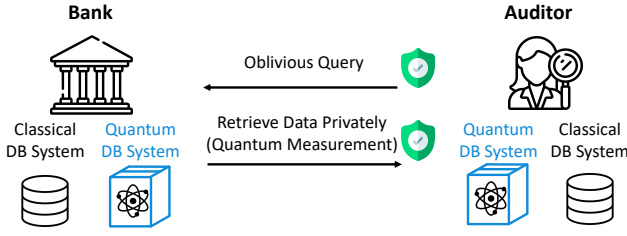


Figure 1. Example: Auditing with user and data privacy

photos [10] and live caller-ID lookup [9]. For SPIR, although not in production, recent experiments demonstrate feasibility over quantum-secure networks for private biometric retrieval with formal server-privacy guarantees [112]. We contend that privacy is an immediate, high-impact opportunity for quantum databases.

Problem Statement. We target a *private quantum database system* that enforces both *user privacy* (the server learns nothing about the query) and *data privacy* (the client learns only the requested record). The following example illustrates the setting.

Example 1. An auditor requires data about a customer from a bank’s databases while satisfying two privacy constraints: first, the auditor’s query (which customer is being audited) must remain oblivious to the bank; second, the auditor must learn nothing beyond that customer’s records. Ideally, quantum database systems, used alongside the classical database systems, facilitate privacy-preserving audits without exposing either the auditor’s intent or other clients’ data.

Contributions. We propose a *private quantum database* that realizes restricted queries for PIR/SPIR by encoding tuples as quantum states and leveraging two fundamental facts: (i) measurement is destructive, so reading one basis erases information about the others; and (ii) the *no-cloning theorem* [116] prevents duplicating unknown quantum states. Together, these physics-level constraints let a single server offer user privacy and bounded data privacy without trusted hardware or heavyweight cryptography.

Concretely, our design: (1) maps keys to mutually unbiased bases and encodes blocks via quantum random access code, so that measuring the chosen basis reconstructs only the requested row while making other rows physically inaccessible; (2) quantifies the correctness–privacy trade-off through the copy budget k and majority-vote amplification; and (3) introduces a hybrid quantum–classical architecture that stores classical descriptions of the prepared states inside a standard DBMS, regenerating fresh instances at query time, practical for NISQ hardware and compatible with today’s database systems.

This framework reframes “quantum advantage” for databases around two-sided privacy rather than speedups, and offers a concrete path to application-driven adoption of quantum components in future database systems.

II. PRELIMINARIES

We first provide some quantum background required for understanding the paper. Readers seeking a more detailed exposition may refer to our previous work [40], [67].

A. Quantum Basics

A *quantum bit (qubit)* is the analogue of a classical bit but can be in a *superposition* of the computational basis² states $|0\rangle$ and $|1\rangle$. That is, a general qubit is of the form

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1,$$

with complex amplitudes α, β . For n qubits we use the *tensor product* $|\psi_1\rangle \otimes \cdots \otimes |\psi_n\rangle$, which expands as a linear combination of basis states such as $|1\rangle \otimes |0\rangle \otimes |1\rangle$; we typically drop $\otimes$ and write $|101\rangle$. In the gate-based quantum computation model, computation proceeds by applying unitary matrices U (*quantum gates*) to these states.

B. Quantum Measurement and Mutually Unbiased Bases

Quantum mechanics forbids direct inspection of quantum states; instead, states (e.g., a computed $U|\psi\rangle$) can only be accessed via measurement. For n qubits, we choose a *measurement basis* $V = \{|v_i\rangle\}_{i=0}^{2^n-1}$, a complete set of 2^n orthonormal n -qubit states with $\langle v_i | v_j \rangle = \delta_{ij}$, where $\delta_{ij} = 1$ if $i = j$ and 0 otherwise. A projective measurement in V collapses $|\psi\rangle$ to $|v_i\rangle$ with probability $p_i = |\langle v_i | \psi \rangle|^2$, and the “measurement apparatus” records the outcome i .

Two orthonormal measurement bases $V = \{|v_i\rangle\}$ and $W = \{|w_j\rangle\}$ on n qubits (so $i, j \in \{0, \dots, 2^n - 1\}$) are *mutually unbiased* if measuring a state prepared in one basis yields a *uniform* outcome distribution in the other: $|\langle v_i | w_j \rangle|^2 = \frac{1}{2^n}$ for all i, j . An n -qubit system has *dimension* (also called the number of *levels*) $d = 2^n$, and admits at most $d+1 = 2^n+1$ MUBs. More generally, a d -level system with d a prime power admits at most $d+1$ MUBs (and such complete sets are known to exist) [115].

C. Quantum Random Access Codes

An $m \rightarrow n$ quantum random access code (QRAC) encodes m classical bits into an n -qubit quantum system, with $m > n$. The goal is to maximize the probability of recovering any chosen bit by performing measurements on the quantum system. In high-dimensional settings ($d = 2^n$), optimal QRACs exploit all $d+1$ MUBs available for prime-power dimensions by preparing a quantum state $|\psi\rangle$ maximizing the overlaps $(\langle v_r | \psi \rangle > 1/d)$ with the designated target basis states $|v_r\rangle$, one chosen in each MUB. This balances the measurement biases across all retrieval choices to an extent, and maximizes the average success probability. [7]

As an example, consider a $3 \rightarrow 1$ QRAC. For a single qubit, the three MUBs are the eigenbases of the *Pauli* observables³ X , Y , and Z . Consider encoding the bit string ‘000’. We

²Column-vector notation: $|0\rangle = [1 \ 0]^T$ and $|1\rangle = [0 \ 1]^T$.

³The Pauli observables are X , Y , and Z , i.e., spin measurements along the Bloch x, y, z axes, with matrices $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

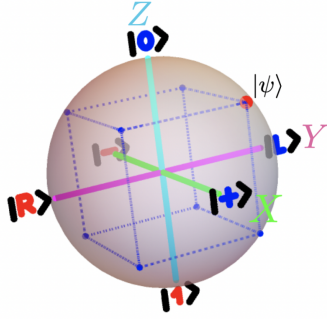


Figure 2. Bloch-sphere representation of $3 \rightarrow 1$ QRAC, which encodes 3 bits in a single qubit, and allows retrieval of any one of them with high probability. All possible single-qubit quantum states lie on the surface of the sphere

choose a state $|\psi\rangle$ that lies midway (on the Bloch sphere; see Fig. 2) between the $+1$ eigenvectors of X , Y , and Z , that is, an intermediate state between $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $|L\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$, and $|0\rangle$. This can be visualized as one corner of a cube embedded within the Bloch sphere, shown as the red point (see Fig. 2). It exhibits a bias towards $|+\rangle$ when measured in the X basis, towards $|L\rangle$ when measured in the Y basis, and towards $|0\rangle$ when measured in the Z basis. Specifically, these desired outcomes occur with probability

$$P_{\text{succ}} = \frac{1 + 1/\sqrt{3}}{2} \approx 0.789,$$

the *success probability* of the $3 \rightarrow 1$ QRAC for any of the three retrieval choices. Other codewords (e.g., xyz with $x, y, z \in \{0, 1\}$) correspond to flipping the Bloch-vector signs along the respective axes, yielding the other seven cube corners.

III. PRIVATE AND RESTRICTED DATABASE QUERIES

We introduce *quantum symmetric private information retrieval* (quantum SPIR), a cryptographic task in which a user retrieves a record from a server-held database while hiding the query (*user privacy*, as in PIR [66]) and, simultaneously, learning nothing beyond the requested record (*data privacy*, as in SPIR [32]). Following the standard in quantum information, we formalize our solution as a protocol that specifies the quantum-classical message flow and proves privacy and correctness probabilistically.

In our setting, the parties (user and server) are quantum-capable and may exchange qubits, prepare quantum states, apply gates, and perform measurements over quantum channels [62]. Conceptually, the classical database is transiently encoded into quantum states, with each data key mapped to a distinct measurement basis. The server prepares and sends a bounded number of quantum state copies⁴, and the user measures in the basis corresponding to the desired key. Because quantum measurement is destructive and incompatible bases disturb the state, attempting to access many items reduces per-item success, creating an intrinsic trade-off between correctness and the amount of information extracted. This trade-off, together with the basis design, yields user

privacy and enforces restricted access for data privacy. As such, the proposed *quantum database* is a classical database with quantum-enhanced query functionalities, providing full user privacy and setting an inverse proportionality between correctness and query size, which grants partial data privacy. We first provide a detailed example in Sec. III-A; the general protocol is presented in Sec. III-B. Correctness and system considerations are discussed in Sec. III-C and we conclude in Sec. III-D with simulation results.

A. Introductory Example

We provide an example of our protocol featuring Alice (the server/database owner) and Bob (the client/querier).

Example 2 Alice has a database containing personal information of 5 users:

Full Name	Phone	Email
Anton Brown	(+31) 123-4567	anton.brown@gmail.com
Brian Smith	(+42) 234-5678	bsmith@yahoo.com
Cynthia Lee	(+51) 345-6789	cynthia.lee@outlook.com
Daniel Kim	(+64) 456-7890	dkim@icloud.com
Eva Martín	(+34) 567-8901	eva.martin@aol.com

For better understandability, we use the full name as the public identifier (key) in this example. Assume each phone number can be represented as a 30-bit integer, and each email as 100 bits by using 11 characters of 8-bit ASCII for the local part and 12 bits to index one of 4096 common domains. Thus, the private payload per user (phone+email) is 130 bits. Here, we omit the full name from this count since the key is known to both parties and is not part of the protected payload.

Bob wishes to learn the information of exactly one user without Alice discovering which one, and Alice does not wish to reveal more of her database than necessary. For simplicity, we assume the values in the *Full Name* column are unique and known to both Alice and Bob; they act as the lookup keys.

We will employ systems of $n = 2$ qubits, which admit $d + 1 = 5$ mutually unbiased bases (MUBs) since $d = 2^n = 4$. Conveniently, this matches the number of database rows ($R = 5$). Because each data row has $C = 130$ bits, Alice prepares 65 independently generated quantum states $|\psi^\ell\rangle$ (i.e., performs 65 two-bit-per-MUB QRAC encodings), where $\ell = 0, 1, \dots, 64$. The state with index ℓ encodes bit positions 2ℓ and $2\ell+1$ for *all* users. Once Bob reliably recovers 130 bits for a single user, he can reconstruct that user's payload. Alternatively, Bob could attempt to recover two users' payloads, but then he must split his limited supply of state instances across two measurement bases, lowering his overall fidelity due to the destructive nature of quantum measurement.

To illustrate, consider the state $|\psi^{18}\rangle$, which encodes the last two bits of the *first* character of each email's ASCII code, namely the final two bits of a, b, c, d, e. These

⁴Preparing copies of *known* quantum states does not contradict the No-Cloning Theorem [88], which applies to unknown states.

are 01, 10, 11, 00, and 01, respectively. Alice uses the following 5 MUBs for two qubits:

MUB #	Commuting observables	Description
1	$\{Z \otimes I, I \otimes Z\}$	Computational (product) basis
2	$\{X \otimes I, I \otimes X\}$	X basis on both qubits
3	$\{Y \otimes I, I \otimes Y\}$	Y basis on both qubits
4	$\{X \otimes Y, Y \otimes Z\}$	Entangled basis
5	$\{Y \otimes X, Z \otimes Y\}$	Entangled basis

Each row specifies one measurement Bob can perform, corresponding to one of the five users he may choose to read. Each such measurement has 4 outcomes 00, 01, 10, 11, each associated with a basis state. To encode the target two-bit strings (01, 10, 11, 00, 01) across the five bases, Alice prepares a single *intermediate* state whose Bloch representation is equidistantly biased toward the desired outcome state in each MUB (the standard QRAC construction). She then sends k independently prepared copies of this state, denoted $|\psi^{18}\rangle$, to Bob.

Bob chooses MUB #1 and measures all k copies in that basis. From known $2 \rightarrow 1$ QRAC performance [23], each copy yields the correct two-bit outcome (here, 01) with average probability 0.5424, and each of the other three outcomes with average probability 0.1525. With k i.i.d. trials, the probability that majority vote returns the correct outcome follows the corresponding multinomial distribution.

In the table below, we report the average success probability $p(M)$ to correctly reconstruct *one* full payload (130 bits) when attempting to read $M \in \{1, 2\}$ users in total, as a function of the number k of copies per encoded two-bit block:

k	$p(1)$	$p(2)$	k	$p(1)$	$p(2)$
1	$\sim 10^{-18}$	$\sim 10^{-18}$	41	0.8927	0.1524
11	0.0006	$\sim 10^{-8}$	51	0.9709	0.3904
21	0.1724	0.0004	61	0.9923	0.6227
31	0.6429	0.0222	71	0.9980	0.7869

Notably, certain values of k (e.g., $k = 41$) yield a strong gap between the success probabilities for retrieving one vs. two users, enhancing data privacy: it is difficult for Bob to recover more than a single payload reliably. To quantify this, define the expected number of successfully retrieved rows $E_r(M) = M p(M)$ and the *data-privacy index* $\mathcal{P}_i = E_r(i)/E_r(i+1)$. When $\mathcal{P}_i > 1$, attempting $i+1$ rows produces *less* total information than attempting i rows.

k	$E_r(1)$	$E_r(2)$	$\mathcal{P}_1$	k	$E_r(1)$	$E_r(2)$	$\mathcal{P}_1$
1	$\sim 10^{-18}$	$\sim 10^{-18}$	1	41	0.8927	0.3048	2.93
11	0.0006	$\sim 10^{-8}$	$\sim 10^4$	51	0.9709	0.7808	1.24
21	0.1724	0.0008	229.5	61	0.9923	1.2455	0.80
31	0.6429	0.0444	14.49	71	0.9980	1.5738	0.63

To construct a protocol where Bob is strongly incentivized to query at most i rows, one can choose the largest k with $\mathcal{P}_i > 1$ (or $\mathcal{P}_i > 2$ for stronger privacy). In this

example, this occurs around $k = 51$ (or $k = 41$). To incentivize exactly $i > 1$ rows, choose k such that $\mathcal{P}_i > 1$ but $\mathcal{P}_{i-1} < 1$; in that case, computing $p(3)$ would be required.

B. The General Protocol

We now present our protocol in full generality.

- Setup (keys $\leftrightarrow$ MUBs).** Alice holds a classical database of size $R \times C$, where R is the number of rows (keys) and C is the number of *binary columns* (bits) per row to be protected. Bob only knows the public keys (e.g., unique names). Alice and Bob agree in advance on a bijection between the R keys and a list of n -qubit MUBs, so that each key/row is associated with one MUB. This maps each classical row to *one* quantum measurement basis, avoiding redundant encodings. The number of rows must satisfy $R \leq 2^n + 1$ to ensure enough MUBs (since $d = 2^n$ admits at most $d+1$ MUBs).
- Blockwise QRAC encoding.** Partition each row's C bits into $L = \lceil C/n \rceil$ contiguous blocks of n bits. For block index $\ell \in \{0, \dots, L-1\}$, let $b_{r,\ell} \in \{0, 1\}^n$ denote the n -bit block of row r . In the MUB assigned to row r , let $|\psi_{b_{r,\ell}}^{(r)}\rangle$ be the basis state labeled by $b_{r,\ell}$. We *encode* block ℓ by constructing an n -qubit state $|\psi^\ell\rangle^5$ that is (QRAC-) biased⁶ toward the R target states $\{|\psi_{b_{r,\ell}}^{(r)}\rangle\}_{r=1}^R$, i.e., it equalizes (and maximizes) the success probability across the R chosen MUBs for that block. Alice stores a *classical description* of how to prepare each $|\psi^\ell\rangle$ (e.g., circuit parameters); no quantum storage is required.
- Private query request and rate limit k .** Bob requests a restricted, private query. Alice grants the request and sets a per-block sending budget of k *independently prepared* instances of each state $|\psi^\ell\rangle$. This does *not* invoke cloning of an unknown state; Alice prepares fresh instances from the known classical description.
- Transmission.** For each block ℓ , Alice prepares and sends k instances of $|\psi^\ell\rangle$ to Bob over a quantum channel and, over a classical channel, communicates the block index ℓ (only). Sending ℓ reveals no row information and is required so Bob can group the received states by block. Alice must make use of an appropriate quantum hardware and quantum circuit setup. (Implementation details of the state preparation and hardware are discussed in Sec. IV.)
- Bob's measurement choices.** Bob selects $M \in \{1, \dots, R\}$ MUBs (thus M keys/rows) to attempt to read. For each block ℓ , he divides the k received instances among the M chosen MUBs (ideally using an odd number per MUB for majority voting) and measures each instance in the corresponding basis, recording the n -bit outcomes.
- Reconstruction.** For each chosen row and each block ℓ , Bob applies majority vote over the outcomes measured in that

⁵Here we mean pure states. A pure state is a quantum state with complete information. It can be described by a single state vector rather than a mixture of several possible states. No classical randomness is involved.

⁶An equidistant bias is desirable for our protocol, but not optimal for QRACs of more than one qubit.

row's MUB to produce an n -bit estimate $\hat{b}_{r,\ell}$. Concatenating the blocks yields an estimated row $\hat{x}_r \in \{0, 1\}^C$. He does so with success probability inversely dependent on M : the number of database keys he attempted to query.

C. Protocol Guarantees & Systems Considerations

We next quantify the success-probability of the quantum query protocol, make explicit the prior agreements between Alice and Bob, and discuss practical feasibility and scalability. **Success probability.** Given a QRAC that uses an n -qubit state to encode n bits in each of up to $2^n + 1$ MUBs, the average success probability of retrieving the n bits correctly can be estimated. This number is known for small QRACs, such as the $n = 1, 2, 3$ QRACs, which have respective success probabilities $p = 0.789, 0.5424, 0.3372$ [23]. Note that this probability can be as low as $1/2^n$, since it attempts to guess n bits. This success probability is amplified by means of majority voting of k quantum-state copies, which we estimate with multinomial distributions. We simplify these distributions by assuming that all unsuccessful outcomes have the same probability, such that the probability of obtaining k_j counts for each bitstring configuration $j = 1, 2, \dots, 2^n$ when measuring the quantum state k times, where $k = \sum_j k_j$, is given by

$$P(k_1, k_2, \dots, k_{2^n}) = \frac{N!}{k_1! k_2! \dots k_{2^n}!} p^{k_1} \left(\frac{1-p}{2^n-1} \right)^{N-k_1}. \quad (1)$$

We employ this expression to approximate the probability that the correct outcome is the one with most counts. This is the process we refer to as *majority voting*.

Prior agreements. Before any query is executed, Alice (data owner) and Bob (client) agree on:

- The database shape $R \times C$. This shape enforces the use of R -MUB encodings in n -qubit, using the smallest n such that $R \leq 2^n + 1$. This shape also defines the number of required encodings to be $\lceil C/n \rceil$.
- a bijection between the R keys and a list of n -qubit mutually unbiased bases (MUBs), with ordered observables. Ordering the observables inside of each MUB is necessary for measurements to yield bitstrings.
- Optionally, the budget of quantum state copies k to send. This number is decided by Alice, but it is convenient to communicate it to Bob, so that he can make an informed decision on what query size and correctness to aim for by planning an appropriate measurement scheme.

Note that these prior agreements only depend on the $R \times C$ database shape. Specifically, they do not depend on the database contents, nor do they depend on Bob's query choices.

Technological feasibility. We argue that the protocol is within reach in practice.

- Quantum enhanced querying is not focused on providing a speedup, but rather two-way privacy, so it does not require quick loading of the classical information into quantum states, a common caveat of most quantum algorithms.
- Random noise in the preparation and measurement of quantum states only changes the baseline QRAC success

Table I
PRELIMINARY SIMULATION RESULTS

MUB #	Expected values	Success probability p
1	+0.620, +0.628	0.6592
2	-0.299, -0.307	0.4244
3	-0.365, -0.345	0.4591
4	-0.330, -0.630	0.5420
5	-0.308, -0.607	0.5255

probability, which simply shifts the required number of copies k for a given correctness and query size. That is, random decoherence in the quantum channel can be corrected by simply increasing a parameter in the protocol.

- A proof of concept of the quantum database query can be performed in superconducting circuits, but a real implementation requires a quantum channel between two parties, which is better suited to single-photon setups.
- In terms of scalability, encoding a modern square database of 3-million rows only requires to prepare and measure 22-qubit quantum states, which is feasible within the limits of current quantum technology. The current challenge to implement a quantum query of this scale lies in computing the states of 22-qubit QRACs (employing all available MUBs), which we reckon should be possible, but demands a dedicated future line of work.

D. Preliminary Simulation Results

As a proof of concept, we have conducted an experiment to validate our proposed quantum protocol. We have implemented the protocol for a simple 2-qubit QRAC case and simulated it using Wolfram Mathematica⁷. The code is available online⁸, and Table I reports the results.

Implementation and experimental setup. We have encoded a 5×2 binary (+1s or -1s) table using 2-qubit quantum states, choosing MUB 1 to correspond to observable values +1, +1 and choosing MUBs 2, 3, 4, 5 to correspond to observable values -1, -1, following the order of the table in the previous example. We numerically computed complex amplitudes a_i for a quantum state of the form $|\psi\rangle = a_1 |00\rangle + a_2 |01\rangle + a_3 |10\rangle + a_4 |11\rangle$ so as to make the $Z \otimes I$ and $I \otimes Z$ expected values as close to +1 as possible, and make the rest as close as possible to -1, via least squares optimization. This simulation was performed with stochastic Hilbert-space search in Mathematica, maximizing the desired observable expected values and decreasing the search space at each iteration.

Result analysis. Up to a global phase, we obtain $a_1 = 0.8162$, $a_2 = -0.2424 - 0.2917i$, $a_3 = -0.2340 - 0.3047i$, $a_4 = 0.0645 - 0.1955i$, yielding the expected values and success probabilities in Table I. For a Pauli observable with expectation E , the probability of outcome +1 is $(E + 1)/2$; we compute the per-MUB success by multiplying the two corresponding probabilities. The mean success across MUBs is 0.5220, consistent with known 2-qubit QRAC behavior [23]. Shared

⁷<https://www.wolfram.com/mathematica/>

⁸<https://doi.org/10.5281/zenodo.17568665>

randomness between Alice and Bob can equalize row-wise success by randomizing key–basis assignments.

The simulation confirms that a 2-qubit QRAC/MUB encoding enables the retrieval with an average success of approximately 0.52 (best-case approximately 0.66), aligning with previous results and guiding the choice of copy budget k for majority-vote amplification in our quantum database.

IV. HYBRID APPROACH

In this section, we show how to integrate the quantum database consisting of quantum states encoding data with classical relational databases.

A. Quantum States in RDBMS

In our quantum SPIR protocol (Sec. III-B), Alice must send, for each block $\ell \in \{0, \dots, L-1\}$ with $L = \lceil C/n \rceil$, up to k independently prepared instances of a known pure state $|\psi^\ell\rangle$ to Bob. On NISQ hardware, *recompiling* and *recalibrating* the state-preparation circuit for every run is costly. Instead, Alice precomputes a *classical description* of each state (or of its preparation circuit) and stores it in a relational DBMS. At query time, she (re)generates fresh instances from this description. To this aim, we need to store, for each block ℓ : (i) the mapping from database key/row r to its MUB index (fixed, see Sec. III-B); (ii) the per-row target basis label $b_{r,\ell} \in \{0, 1\}^n$ (the n bits of row r that block ℓ encodes); (iii) a classical description of the *intermediate* QRAC state $|\psi^\ell\rangle$ (or of a circuit U_ℓ such that $U_\ell |0\rangle^{\otimes n} = |\psi^\ell\rangle$). Items (i)–(iii) are classical and fit naturally in tables; they let Alice recreate $|\psi^\ell\rangle$ on demand and dispatch k fresh instances when Bob initiates a query.

We consider standard classical representations of quantum states used in simulation and compilation pipelines: state vectors [88], tensor networks [16], decision diagrams [114], and the stabilizer tableau [1]. Our existing system, Qymera [75], already supports state vectors, tensor networks, and decision diagrams in relational engines such as SQLite and DuckDB. For completeness and because several ingredients of our protocol (e.g., measurements, MUB observables) are stabilizer-heavy, we also support the *tableau* representation inside an RDBMS.

B. Tableau Representation

The n -qubit Pauli group is $\mathcal{P}_n = \{\pm P_1 \otimes \dots \otimes P_n \mid P_j \in \{I, X, Y, Z\}\}$ with I identity and X, Y and Z the Pauli observables encountered earlier. A pure state $|\psi\rangle$ is a *stabilizer state* if there exists a set of n commuting, independent Paulis $\{g_1, \dots, g_n\} \subset \mathcal{P}_n$ such that $g_i |\psi\rangle = |\psi\rangle$ for all i . These n generators generate a group of size 2^n that fixes $|\psi\rangle$ and only $|\psi\rangle$. For example, on two qubits, $|00\rangle$ is stabilized by the group $\{II, ZI, IZ, ZZ\}$, but only *two* commuting generators, say ZI and IZ , are *independent*; $ZZ = (ZI)(IZ)$ and II is the identity. Hence two generators suffice for $n=2$. The de facto compact encoding for stabilizer states is the *tableau* [1], [46]. For n qubits, the tableau stores n commuting generators as a triple $T = (X_M, Z_M, S_M)$ with $X_M, Z_M \in \{0, 1\}^{n \times n}$ and $S_M \in \{0, 1\}^n$:

- Row i encodes generator g_i ; column j encodes the Pauli on qubit j . The pair $(X_M[i, j], Z_M[i, j])$ decodes to I, X, Z, Y as $(0, 0), (1, 0), (0, 1), (1, 1)$.
- The sign bit $S_M[i] = 0 (+)$ or $1 (-)$ records the overall phase of g_i .

Example 3 (tableau for two generators). Generators of the stabilizer group of $|00\rangle$ are encoded as

$$T = \left[\begin{array}{cc|cc|c} X_M & Z_M & S_M \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right],$$

where indeed the first row encodes $-ZI$, the second IZ .

Stabilizers can be stored and manipulated *compactly* as tableaux and updated by bitwise row operations in $O(n)$ time, ideal for an RDBMS backend. When a block state $|\psi^\ell\rangle$ is non-stabilizer (typical for QRAC intermediates), we store its *circuit parameters*, to regenerate it later on hardware.

We record every Pauli symbol with two Boolean *flags*: the first flag stands for the presence of X , the second for Z (Tab. II). Concatenating the $2n$ flags yields a fixed-width bit string that is compact yet easy to manipulate inside an SQL engine. Below we illustrate one possible schema definition

Table II
TWO-FLAG CODING OF A SINGLE-QUBIT PAULI OPERATOR

Flags (x, z)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
Pauli operator	I	X	Z	Y

using PostgreSQL syntax; other relational engines that support a fixed-length BIT datatype can be used in a similar fashion. Based on this schema, standard SQL queries can be written to retrieve and manipulate qubit-related information.

Listing 1. Schema for storing Pauli generators.

```
CREATE TABLE generators (
  circ_id BIGINT,      -- circuit ID
  g_idx SMALLINT,      -- generator index
  n SMALLINT,          -- number of qubits
  flags VARBIT NOT NULL, -- 2n bits = Pauli string
  neg BOOLEAN,         -- TRUE = overall minus sign
  PRIMARY KEY (circ_id, g_idx),
  CHECK (bit_length(flags) = 2 * n)
);
```

C. Hybrid Quantum-Classical Approach

Fig. 3 illustrates our hybrid approach. Classical data stored in a classical database system (e.g., an RDBMS) serves as the input to the quantum database system. On the *server* side (Alice), the system consists of three components: (i) given a classical database with size of $R \times C$, *Key–Basis Mapping*, which assigns each primary key to a mutually unbiased basis; (ii) the *Quantum Encoder (QRAC)*, which transforms the classical data into n -qubit states; and (iii) *Qubit Preparation*, which instantiates these states on quantum hardware and transmits k copies to the client.

On the *client* side (Bob), retrieving the requested data involves two components. The *Measurement & Basis Selection* component chooses up to M measurement bases

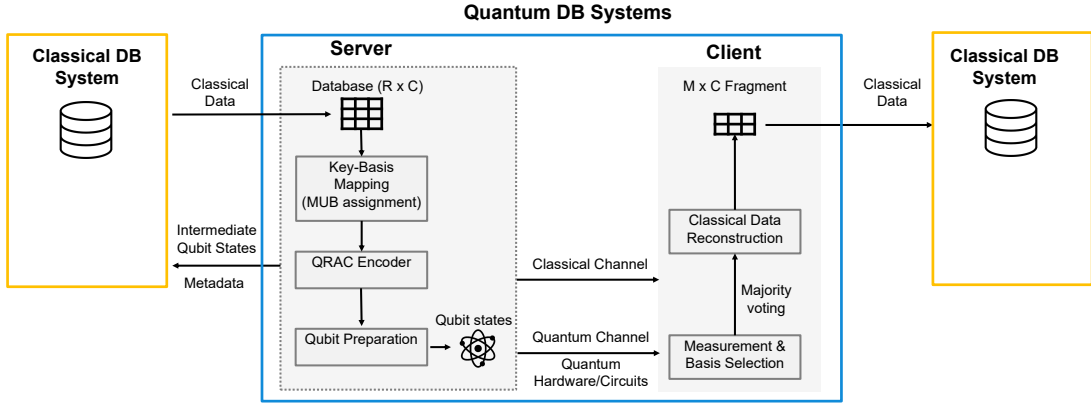


Figure 3. Overview of our hybrid quantum-classical approach

and performs measurements on the received qubits. The resulting outcomes are locally processed via majority voting and reconstructed by the *Classical Data Reconstruction* component into an $M \times C$ fragment of the classical database, which can then be stored in a classical DBMS.

The role of classical database systems (e.g., RDBMSs) includes storing the server- and client-side classical data, the classical descriptions of the intermediate QRAC states from which the transmitted qubits are prepared, together with the metadata that drives our protocol, namely the mapping from each key/row r to its assigned MUB index and the per-row target basis labels.

D. Ongoing Implementation

We consider our example scenario (Sec. III-A) as a starting point. The quantum states sent by Alice in our database query protocol can be generated with the general quantum circuit in Fig. 4, capable of preparing any 2-qubit quantum state [88]. We also include corresponding quantum circuits for Bob to measure in any of the 5 MUBs of 2 qubits. Note that Alice prepares intermediate states between MUB elements, so there are $4^5 = 1024$ states to choose from (4 options per MUB), whereas Bob only requires to choose one of 5 MUBs to perform measurements.

Superconducting circuits via cloud quantum computing are an appropriate hardware to test the quantum-state generation and measurement in our protocol. In particular, we plan to conduct experiments on the superconducting processors *Tuna-5* and *Starmon-7*, available through the Quantum Inspire infrastructure developed at TU Delft⁹. Moreover, to implement a working proof of principle where Alice and Bob are in different locations, photonic systems are a more natural fit due to their suitability for long-distance quantum communication. To this end, we can either employ two-photon systems, encoding each of the two qubits of the protocol in their polarization degree of freedom, or we can alternatively employ single-photon systems, encoding one qubit in the polarization degree of freedom and the other in the spatial-mode degree of freedom, as done in [79].

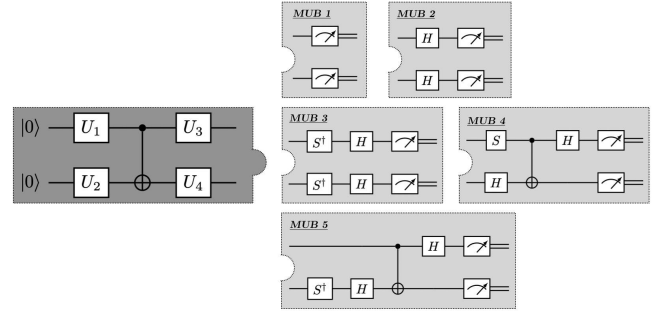


Figure 4. Quantum circuits for Alice to prepare any 2-qubit state (dark grey) and for Bob to measure any 2-qubit MUB (light grey). Each horizontal line represents a qubit that is transformed with quantum gates. The boxed U_i , S , $S^\dagger$ and H are single-qubit gates typically implemented in photonics via waveplates, and the dot-line-circle symbol is a Controlled-NOT gate, typically implemented in photonics via interferometer setups.

Moreover, we are extending our system Qymera [75], which stores the data representations of quantum states as relational tables and compiles quantum gate operations and measurement computations into SQL queries executed on RDBMS engines such as SQLite and DuckDB. As introduced in Sec. IV-B, we are now adding support for the tableau representation for stabilizer states. In parallel, we are developing a quantum database optimizer that (i) selects appropriate quantum states, (ii) tunes performance-related parameters such as M and k , and (iii) balances workloads between quantum and classical execution.

V. RELATED WORK

This work lies at the intersection of data management, quantum computing, and privacy and security. To position our contribution, this section reviews prior quantum-related efforts within the database community (Sec. V-A–V-B), surveys existing work on classical PIR, SPIR, and their quantum variant QSPIR (Sec. V-C–V-D), and relates our approach to other privacy and security techniques (Sec. V-E).

A. Classical DB & Quantum Computing

Recent quantum-focused efforts within the DB community can be divided into two directions: first, leveraging quantum computers as new hardware to boost classical database tasks,

⁹<https://www.quantum-inspire.com/backends/transmons/>

such as query optimization [100], [101], [108], data integration [41], index selection [63], and transaction management [48]. The other direction is to use database technologies for quantum computing [53], such as utilizing RDBMSs for running quantum circuit simulation [19], [75], [107].

Our work aligns primarily with the first research direction: we leverage quantum computing techniques (i.e., QRAC, MUBs) to design a quantum-native database system that supports symmetric private information retrieval. Meanwhile, our implementation engages with the second research direction: Sec. IV explains how we manage classical representations of quantum states within an RDBMS, building on our earlier efforts that use relational databases for quantum computing.

B. Quantum DBMSs Based on Grover’s Algorithm

A *database* is a persistent, electronically stored collection of data [2], [94]. In the quantum community, the term *database*¹⁰ usually denotes a conceptual framework for processing and searching data using quantum algorithms. Grover’s algorithm searches an unsorted set of N items in $\mathcal{O}(\sqrt{N})$ iterations instead of the classical $\mathcal{O}(N)$ bound [49]. During the 1990s–2010s, a series of works extended the basic idea of Grover’s algorithm to searching databases [21], [56], [59], [90], [106], [109]. Moreover, several attempts have been made to build full relational “quantum DBMSs” in which SQL-like statements are realized by quantum circuits, supporting relational operations such as joins and set operations (intersection, union, difference) [28], [50], [51], [58], [89], [99], and data manipulation operations (insert, update, delete) [50], [51], [117].

However, these approaches do not translate into practice in terms of algorithmic cost and in physical implementability. First, each query contains a quantum measurement, which destroys the qubit state, so the qubit state must be rebuilt for the next query, resulting in $\mathcal{O}(N)$ time, canceling Grover’s $\sqrt{N}$ speed-up. Second, NISQ quantum computers are noisy and limited in qubits. The circuit depths and number of qubits required in the above proposals, even for small relational tables, are beyond the capabilities of NISQ hardware. Thus, in this work, we do *not* pursue Grover-style quantum databases. We prioritize quantum techniques that are technologically feasible given current constraints on qubit count and noise. As discussed in Sec. III-C, using the schema in Example 2, a database with 3 million rows requires only 22 qubits, which is achievable with today’s NISQ quantum processors.

C. Classical PIR and SPIR

Private information retrieval (PIR) was first discussed in [27], [66]. The main challenges typically fall into two categories: designing communication-efficient approaches [22], [25], [33], [74] and computation-efficient approaches [5], [8], [83]. To further reduce computation cost, recent works apply fully homomorphic encryption [6], [84], [87] or perform partial computation in advance [54]. Recent work [68] also

supports complicated scenarios with skewed query distributions, where some servers receive more queries than others.

PIR has been studied in several concrete data management settings. Ghinita et al. [43] use PIR techniques for location-based services, enabling private location queries without trusted anonymizers. Pantheon [3] cuts single-round, server-side latency in key-value stores to two million item lookups in about one second. The tutorial [4] surveys both theoretical developments and practical engines for PIR over public datasets. S^2 [102] extends multi-server, information-theoretic PIR and supports keyword search and retrieval of rows obliviously over relational tables. Femur [119] lets clients trade bounded privacy loss for better performance in public key-value stores.

Symmetric private information retrieval (SPIR) provides *two-sided* privacy, both data and user privacy, preventing either party from learning more than necessary [42]. In the classical, information-theoretic setting, SPIR is achievable only with *multiple* servers that share common randomness (a shared random string) unknown to the user [42]. Recent work, XSPIR [73], presents an efficient Ring-LWE-based SPIR scheme that leverages homomorphic encryption to provide both user and database privacy, achieving performance comparable to leading PIR systems. However, classical SPIR solutions generally remain communication- and computation-heavy.

In contrast, quantum communication can, in principle, achieve SPIR with a single server. Our proposed quantum database tries to achieve two-sided privacy via quantum QRAC/MUB encoding and destructive measurement.

D. Quantum SPIR

For completeness, we briefly outline recent advances in quantum protocols to contextualize our work within the broader quantum landscape. Readers focused solely on classical database systems may safely skip this section.

In the *multi-server* setting, quantum protocols enable sublinear communication *without shared randomness*, which is unachievable in classical SPIR [62]. For the *single-server* case, Quantum private query (QPQ) addresses symmetric PIR in the quantum regime. QPQ offers *cheat-sensitive*, two-sided privacy with exponential gains in communication and computation over classical SPIR protocols [44], [45]. It employs two quantum states: one state for encoding the query, and a superposition decoy state to detect server-side cheating, while accessing the database via a quantum oracle. The server returns two items using only $\mathcal{O}(\log N)$ qubits, avoiding the need to send the full $\mathcal{O}(N)$ database, where N is the database size. A proof-of-concept experiment [30] employed linear optics, using photon momentum and polarization as address and bus qubits.

Due to challenges in scaling the dimension of the oracle operation for large databases in the original QPQ, more practical variants have been explored. A *loss-tolerant* QPQ protocol based on quantum key distribution (QKD) was demonstrated in [57], with communication-efficient extensions proposed in [38]. Other variants include high-dimensional

¹⁰In [98], the authors use the term “quantum database” purely as a metaphor in a classical resource-allocation context. Their work is unrelated to quantum computing, and falls outside the scope of this paper.

QPQ supporting multi-bit retrieval [113], counterfactual QKD-based QPQ [118], and passive round-robin differential phase-shift QKD implementations [69]. An implementation over a 12.4 km fiber link incorporating error correction was demonstrated in [24].

Recent work further refines practicality and privacy. A protocol ensuring zero failure probability and bounded database information leakage is introduced in [76]. Decoy-state methods resilient to loss and noise were proposed in [77], and extended in [93] to mitigate multi-photon attacks.

From a theoretical standpoint, a relativistic QPQ protocol leveraging Minkowski causality was proposed in [103]. A *device-independent* QPQ approach using CHSH tests [72] was introduced in [80], removing trust assumptions on the quantum devices.

In summary, first, most single-server QSPIR approaches are built on quantum private query [44]. Instead, our private quantum database employs quantum encoding (i.e., QRACs over mutually unbiased bases), which is a new type of protocol not previously explored for the QSPIR setting. Second, QPQ provides cheat-sensitive user privacy and only bounded (i.e., non-perfect) data privacy. In our design, data privacy is parameter-bounded (e.g., by the copy count k) and thus imperfect, whereas user privacy is ensured natively by the protocol.

E. Complementary to Data Confidentiality and MPC

Our approach is *complementary* to two established privacy and security techniques: data confidentiality and secure multi-party computation (MPC). Our focus is on single-server SPIR, which ensures both user privacy (the server learns nothing about the query) and data privacy (the client learns only the requested record). In contrast, data confidentiality addresses a different problem, aiming to protect data from the server itself. Similarly, MPC enables joint computation among mutually distrustful parties. Although these techniques tackle different problems from ours, our approach can be integrated with them to form a broader privacy-preserving data architecture.

Data Confidentiality. Existing encrypted databases focus on data confidentiality: when users move their workload to the cloud, they want to keep their data confidential to the cloud, i.e., the DBMS on the server executes queries without revealing unnecessary information. Hacigümüs et al. [52] tackle the challenge of maintaining data confidentiality on untrusted cloud providers and propose processing SQL queries over encrypted cloud databases. MONOMI [110] splits analytical queries between an encrypted cloud server and an unencrypted client engine for higher efficiency. Performance can be further improved with secure hardware, such as IBM cryptographic coprocessors in TrustedDB [12]. Many systems rely on homomorphic encryption [18], [96], [121], often causing high computation overhead. Arx [92] takes a different path, using only semantically secure encryption schemes. More recent works employ differential privacy for private SQL processing [65], or formalize leakage-safe design rules across heterogeneous cryptographic primitives [120].

Another promising technology for improving the security of databases outsourced to the cloud is secure enclaves, which are a hardware-based isolation mechanism recently introduced in modern CPUs such as Intel SGX [82] and AMD Memory Encryption [61]. To guarantee the access-pattern privacy (i.e., hiding which tuples, blocks, or operators are touched), systems are built to support data storage [104], query processing [11], [36], [122], ACID transactions [29], search index [85], and fault tolerant [81] in an oblivious manner.

Secure multi-party computation. Beyond single-server PIR, compiler-based and MPC-style frameworks achieve collaborative analytics among multiple distrustful organizations. SM-CQL [13] tackles the problem of processing and optimizing secure SQL queries over a federated database with mutually distrustful data owners. The queries are sent and executed by an honest broker, and thus are oblivious to others except the broker and the querier. To scale relational analytics queries, Conclave [111] serves as a query compiler that enables a hybrid execution pipeline combining cleartext processing with secure multiparty computation, minimizing the computational overhead incurred by MPC operations. Secure multiparty computation is employed for collaborative analytics on secret-shared data in the cloud: SECRECY [71] introduces oblivious relational operators and a cost model, while SCQL [37] implements an industrial-grade secure query platform. Our proposed quantum database can be integrated into existing MPC frameworks, for example, by serving as a private retrieval step.

VI. CONCLUSION

We position privacy as one of the most immediate and impactful benefits that quantum computing can offer to future database technologies. We propose a quantum database design that aims at both user and data privacy by encoding relational tuples as quantum random access codes over mutually unbiased bases. Besides this protocol, we outline a hybrid approach that embeds quantum primitives into existing DBMS engines, paving the way toward practical, application-driven quantum database systems.

AI-GENERATED CONTENT ACKNOWLEDGMENT

We used OpenAI ChatGPT to polish the language and improve the readability of certain parts of this manuscript. The tool was employed for stylistic refinement, searching references, and format adjustment. All ideas, technical content, and conclusions were produced and verified by the authors.

REFERENCES

- [1] Scott Aaronson and Daniel Gottesman. Improved simulation of stabilizer circuits. *Physical Review A*, 70(5):052328, 2004.
- [2] Serge Abiteboul, Richard Hull, and Victor Vianu. *Foundations of databases*. Addison-Wesley, 1995.
- [3] Ishtiaque Ahmad, Divyakant Agrawal, Amr El Abbadi, and Trinabh Gupta. Pantheon: Private retrieval from public key-value store. *Proceedings of the VLDB Endowment*, 16(4):643–656, 2022.
- [4] Ishtiaque Ahmad, Divyakant Agrawal, Amr El Abbadi, and Trinabh Gupta. Private information retrieval in large scale public data repositories. *Proceedings of the VLDB Endowment*, 16(12):3868–3871, 2023.

- [5] Ishtiaque Ahmad, Yuntian Yang, Divyakant Agrawal, Amr El Abbadi, and Trinabh Gupta. Addra: Metadata-private voice communication over fully untrusted infrastructure. In *15th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2021.
- [6] Asra Ali, Tancrède Lepoint, Sarvar Patel, Mariana Raykova, Philipp Schoppmann, Karn Seth, and Kevin Yeo. Communication-Computation trade-offs in PIR. In *30th USENIX Security Symposium*, pages 1811–1828, 2021.
- [7] Andris Ambainis, Debbie Leung, Laura Mancinska, and Maris Ozols. Quantum random access codes with shared randomness. In *ArXiv, 0810.2937*, 2009.
- [8] Sebastian Angel, Hao Chen, Kim Laine, and Srinath Setty. Pir with compressed queries and amortized query processing. In *2018 IEEE symposium on security and privacy (SP)*, pages 962–979. IEEE, 2018.
- [9] Apple. Pir service example. <https://github.com/apple/pir-service-example>, October 2025. GitHub repository; commit 1a0747b1dae7325df8b439a9782ceb68ea35e73a (2025-10-01); Apache-2.0 License.
- [10] Apple Machine Learning Research. Combining machine learning and homomorphic encryption in the apple ecosystem. <https://machinelearning.apple.com/research/homomorphic-encryption>, 2024. Published Oct. 24, 2024; accessed Nov. 7, 2025.
- [11] Arvind Arasu, Spyros Blanas, Ken Eguro, Raghav Kaushik, Donald Kossmann, Ravishankar Ramamurthy, and Ramarathnam Venkatesan. Orthogonal security with cipherbase. In *CIDR*, 2013.
- [12] Sumeet Bajaj and Radu Sion. TrustedDB: A trusted hardware based database with privacy and data confidentiality. In Timos K. Sellis, Renée J. Miller, Anastasios Kementsietsidis, and Yannis Velegrakis, editors, *Proceedings of the ACM SIGMOD International Conference on Management of Data (SIGMOD)*, pages 205–216. ACM, 2011.
- [13] Johes Bater, Gregory Elliott, Craig Eggen, Satyender Goel, Abel Kho, and Jennie Rogers. SMCQL: Secure querying for federated databases. *Proceedings of the VLDB Endowment*, 10(6):673–684, 2017.
- [14] Charles H. Bennett and Gilles Brassard. Quantum cryptography: Public key distribution and coin tossing. In *Proceedings of the IEEE International Conference on Computers, Systems and Signal Processing*, pages 175–179. IEEE, December 1984.
- [15] Charles H Bennett, Gilles Brassard, Claude Crépeau, and Marie-Hélène Skubiszewska. Practical quantum oblivious transfer. In *Annual international cryptology conference*, pages 351–366. Springer, 1991.
- [16] Jacob Biamonte. Lectures on quantum tensor networks. In *arXiv, 1912.10049*, 2020.
- [17] Jacob Biamonte, Peter Wittek, Nicola Pancotti, Patrick Rebentrost, Nathan Wiebe, and Seth Lloyd. Quantum machine learning. *Nature*, 549(7671):195–202, 2017.
- [18] Song Bian, Zhou Zhang, Haowen Pan, Ran Mao, Zian Zhao, Yier Jin, and Zhenyu Guan. HE3DB: an efficient and elastic encrypted database via arithmetic-and-logic fully homomorphic encryption. In Weizhi Meng, Christian Damsgaard Jensen, Cas Cremers, and Engin Kirda, editors, *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, pages 2930–2944. ACM, 2023.
- [19] Mark Blacher, Julien Klaus, Christoph Staudt, Sören Laue, Viktor Leis, and Joachim Giesen. Efficient and portable einstein summation in sql. *PACMOD*, 1(2), jun 2023.
- [20] Dolev Bluvstein, Simon J Evered, Alexandra A Geim, Sophie H Li, Hengyun Zhou, Tom Manovitz, Sepehr Ebadi, Madelyn Cain, Marcin Kalinowski, Dominik Hangleiter, et al. Logical quantum processor based on reconfigurable atom arrays. *Nature*, 626(7997):58–65, 2024.
- [21] Michel Boyer, Gilles Brassard, Peter Høyer, and Alain Tapp. Tight bounds on quantum searching. *Fortschritte der Physik: Progress of Physics*, 46(4-5):493–505, 1998.
- [22] Christian Cachin, Silvio Micali, and Markus Stadler. Computationally private information retrieval with polylogarithmic communication. In *International Conference on the Theory and Applications of Cryptographic Techniques*, pages 402–414. Springer, 1999.
- [23] Andrea Casaccino, Ernesto F. Galvão, and Simone Severini. Extrema of discrete wigner functions and applications. *Physical Review A*, 78(2), August 2008.
- [24] P. Chan, I. Lucio-Martinez, X. Mo, C. Simon, and W. Tittel. Performing private database queries in a real-world environment using a quantum protocol. *Scientific Reports*, 4:5233, 2014.
- [25] Melissa Chase, Sanjam Garg, Mohammad Hajiabadi, Jialin Li, and Peihan Miao. Amortizing rate-1 OT and applications to PIR and PSI. In *Theory of Cryptography Conference*, pages 126–156. Springer, 2021.
- [26] Sitan Chen, Jordan Cotler, Hsin-Yuan Huang, and Jerry Li. The complexity of nisq. *Nature Communications*, 14(1):6001, 2023.
- [27] Ben-Zion Chor, Oded Goldreich, and Eyal Kushilevitz. Private information retrieval, December 29 1998. US Patent 5,855,018.
- [28] Paul Cockshott. Quantum relational databases. In *ArXiv, quant-ph, 9712025*, 1997.
- [29] Natacha Crooks, Matthew Burke, Ethan Cecchetti, Sitar Harel, Rachit Agarwal, and Lorenzo Alvisi. Obladi: oblivious serializable transactions in the cloud. In *Proceedings of the 13th USENIX Conference on Operating Systems Design and Implementation*, page 727–743, USA, 2018.
- [30] Francesco De Martini, Vittorio Giovannetti, Seth Lloyd, Lorenzo Maccone, Eleonora Nagali, Linda Sansoni, and Fabio Sciarrino. Experimental quantum private queries with linear optics. *Physical Review A—Atomic, Molecular, and Optical Physics*, 80(1):010302, 2009.
- [31] Matthew DeCross, Reza Haghshenas, Minzhao Liu, Enrico Rinaldi, Johnnie Gray, Yuri Alexeev, Charles H Baldwin, John P Bartolotta, Matthew Bohn, Eli Chertkov, et al. Computational power of random quantum circuits in arbitrary geometries. *Physical Review X*, 15(2):021052, 2025.
- [32] Giovanni Di Crescenzo, Tal Malkin, and Rafail Ostrovsky. Single database private information retrieval implies oblivious transfer. In *International Conference on the Theory and Applications of Cryptographic Techniques*, pages 122–138. Springer, 2000.
- [33] Nico Döttling, Sanjam Garg, Yuval Ishai, Giulio Malavolta, Tamer Mour, and Rafail Ostrovsky. Trapdoor hash functions and their applications. In *Annual International Cryptology Conference*, pages 3–32. Springer, 2019.
- [34] Jens Eisert and John Preskill. Mind the gaps: The fraught road to quantum advantage. In *arXiv, 2510.19928*, 2025.
- [35] Ramez Elmasri and Shamkant B Navathe. *Fundamentals of Database Systems*. Pearson, 2016.
- [36] Saba Eskandarian and Matei Zaharia. Oblidb: Oblivious query processing for secure databases. *Proc. VLDB Endow.*, 13(2):169–183, 2019.
- [37] Wenjing Fang, Shunde Cao, Guojin Hua, Junming Ma, Yongqiang Yu, Qunshan Huang, Jun Feng, Jin Tan, Xiaopeng Zan, Pu Duan, et al. SecretFlow-SCQL: A secure collaborative query platform. *Proceedings of the VLDB Endowment*, 17(12):3987–4000, 2024.
- [38] Fei Gao, Bin Liu, Qiao-Yan Wen, and Hui Chen. Flexible quantum private queries based on quantum key distribution. *Optics Express*, 20(16):17411–17420, 2012.
- [39] Hector Garcia-Molina. *Database systems: The complete book*. Pearson, 2008.
- [40] Giancarlo Gatti, Daniel Huerger, Enrique Solano, and Mikel Sanz. Random access codes via quantum contextual redundancy. *Quantum*, 7:895, 2023.
- [41] Luisa Gerlach, Tobias Köppl, Stefanie Scherzinger, Nicole Schweikardt, and René Zander. A quantum-leap into schema matching: Beyond 1-to-1 matchings. *Proc. ACM Manag. Data*, 3(2), June 2025.
- [42] Yael Gertner, Yuval Ishai, Eyal Kushilevitz, and Tal Malkin. Protecting data privacy in private information retrieval schemes. In *Proceedings of the thirtieth annual ACM symposium on Theory of Computing (STOC)*, pages 151–160, 1998.
- [43] Gabriel Ghinita, Panos Kalnis, Ali Khoshgozaran, Cyrus Shahabi, and Kian-Lee Tan. Private queries in location based services: anonymizers are not necessary. In *Proceedings of the 2008 ACM SIGMOD international conference on Management of data*, pages 121–132, 2008.
- [44] Vittorio Giovannetti, Seth Lloyd, and Lorenzo Maccone. Quantum private queries. *Physical review letters*, 100(23):230502, 2008.
- [45] Vittorio Giovannetti, Seth Lloyd, and Lorenzo Maccone. Quantum private queries: Security analysis. *IEEE Transactions on Information Theory*, 56(7):3465–3477, 2010.
- [46] Daniel Gottesman. Stabilizer codes and quantum error correction, 1997.
- [47] TM Graham, Y Song, J Scott, C Poole, L Phuttitarn, K Jooya, P Eichler, X Jiang, A Marra, B Grinkemeyer, et al. Multi-qubit entanglement and algorithms on a neutral-atom quantum computer. *Nature*, 604(7906):457–462, 2022.
- [48] Sven Groppe and Jinghua Groppe. Optimizing transaction schedules on universal quantum computers via code generation for grover’s search algorithm. In *Proceedings of the 25th International Database*

- Engineering & Applications Symposium (IDEAS)*, pages 149–156, 2021.
- [49] Lov K Grover. A Fast Quantum Mechanical Algorithm for Database Search. In *Proceedings of the Twenty-Eighth Annual ACM Symposium on Theory of Computing (STOC)*, pages 212–219, 1996.
 - [50] Amor Gueddana, Rihab Chatta, and Moez Attia. CNOT-based design and query management in quantum relational databases. *International Journal of Quantum Information*, 12(04):1450023, 2014.
 - [51] Amor Gueddana, Rihab Chatta, and Nouredine Boudriga. Optimized methods for inserting and deleting records and data retrieving in quantum database. In *2010 12th International Conference on Transparent Optical Networks*, pages 1–5. IEEE, 2010.
 - [52] Hakan Hacigümüs, Balakrishna R. Iyer, Chen Li, and Sharad Mehrotra. Executing SQL over encrypted data in the database-service-provider model. In *Proceedings of the 2002 ACM SIGMOD International Conference on Management of Data (SIGMOD)*, pages 216–227. ACM, 2002.
 - [53] Rihan Hai, Shih-Han Hung, Tim Coopmans, Tim Littau, and Floris Geerts. Quantum data management in the NISQ era. *PVLDB*, 18(6):1720–1729, 2025.
 - [54] Alexandra Henzinger, Matthew M Hong, Henry Corrigan-Gibbs, Sarah Meiklejohn, and Vinod Vaikuntanathan. One server for the price of two: Simple and fast Single-Server private information retrieval. In *32nd USENIX Security Symposium (USENIX Security 23)*, pages 3889–3905, 2023.
 - [55] Hsin-Yuan Huang, Michael Broughton, Jordan Cotler, Sitan Chen, Jerry Li, Masoud Mohseni, Hartmut Neven, Ryan Babbush, Richard Kueng, John Preskill, et al. Quantum advantage in learning from experiments. *Science*, 376(6598):1182–1186, 2022.
 - [56] Sándor Imre and Ferenc Balázs. The generalized quantum database search algorithm. *Computing*, 73:245–269, 2004.
 - [57] Markus Jakob, Christoph Simon, Nicolas Gisin, Jean-Daniel Bancal, Cyril Branciard, Nino Walenta, and Hugo Zbinden. Practical private database queries based on a quantum-key-distribution protocol. *Physical Review A—Atomic, Molecular, and Optical Physics*, 83(2):022301, 2011.
 - [58] Szabolcs Józscsik and Attila Kiss. Quantum Computation and Its Effects in Database Systems. In *European Conference on Advances in Databases and Information Systems*, pages 13–23. Springer, 2020.
 - [59] Yi-Lin Ju, I-Ming Tsai, and Sy-Yen Kuo. Quantum circuit design and analysis for database search applications. *IEEE Transactions on Circuits and Systems I: Regular Papers*, 54(11):2552–2563, 2007.
 - [60] V Kalaivani et al. Enhanced bb84 quantum cryptography protocol for secure communication in wireless body sensor networks for medical applications. *Personal and ubiquitous computing*, 27(3):875, 2021.
 - [61] David Kaplan, Jeremy Powell, and Tom Woller. Amd memory encryption. White Paper 70363, Advanced Micro Devices, Inc., Oct 2021.
 - [62] Iordanis Kerenidis and Ronald De Wolf. Quantum symmetrically-private information retrieval. *Information Processing Letters*, 90(3):109–114, 2004.
 - [63] Manish Kesarwani and Jayant R. Haritsa. Index advisors on quantum platforms. *Proceedings of the VLDB Endowment (PVLDB)*, 17(11):3615–3628, 2024.
 - [64] Youngseok Kim, Andrew Eddins, Sajant Anand, Ken Xuan Wei, Ewout Van Den Berg, Sami Rosenblatt, Hasan Nayfeh, Yantao Wu, Michael Zaletel, Kristan Temme, et al. Evidence for the utility of quantum computing before fault tolerance. *Nature*, 618(7965):500–505, 2023.
 - [65] Ios Kotsogiannis, Yuchao Tao, Xi He, Maryam Fanaeepour, Ashwin Machanavajjhala, Michael Hay, and Gerome Miklau. PrivateSQL: a differentially private sql query engine. *Proceedings of the VLDB Endowment*, 12(11):1371–1384, 2019.
 - [66] Eyal Kushilevitz and Rafail Ostrovsky. Replication is not needed: Single database, computationally-private information retrieval. In *Proceedings 38th of the Annual IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 364–373. IEEE, 1997.
 - [67] Jeffrey Lazar, Santiago Giner Olavarrieta, Giancarlo Gatti, Carlos A Argüelles, and Mikel Sanz. Pathways in neutrino physics via quantum-encoded data analysis. *Physical Review Research*, Jul 2025.
 - [68] Ryan Lehmkuhl, Alexandra Henzinger, and Henry Corrigan-Gibbs. Distributional private information retrieval. In *34th USENIX Security Symposium (USENIX Security)*, pages 3377–3396, 2025.
 - [69] Jian Li, Yu-Guang Yang, Xiu-Bo Chen, Yi-Hua Zhou, and Wei-Min Shi. Practical quantum private database queries based on passive round-robin differential phase-shift quantum key distribution. *Scientific Reports*, 6:31738, 2016.
 - [70] Zhiyuan Li, Pei Liu, Peng Zhao, Zhenyu Mi, Huikai Xu, Xuehui Liang, Tang Su, Weijie Sun, Guangming Xue, Jing-Ning Zhang, Weiyang Liu, Yirong Jin, and Haifeng Yu. Error per single-qubit gate below 10⁻⁴ in a superconducting qubit. *npj Quantum Information*, 9(1):111, 2023.
 - [71] John Liagouris, Vasiliki Kalavri, Muhammad Faisal, and Mayank Varia. SECRECY: Secure collaborative analytics in untrusted clouds. In *20th USENIX Symposium on Networked Systems Design and Implementation (NSDI 23)*, pages 1031–1056, 2023.
 - [72] Charles Ci Wen Lim, Christopher Portmann, Marco Tomamichel, Renato Renner, and Nicolas Gisin. Device-independent quantum key distribution with local bell test. *Phys. Rev. X*, 3:031006, Jul 2013.
 - [73] Chengyu Lin, Zeyu Liu, and Tal Malkin. Xspir: Efficient symmetrically private information retrieval from ring-lwe. In *European Symposium on Research in Computer Security*, pages 217–236. Springer, 2022.
 - [74] Helger Lipmaa. An oblivious transfer protocol with log-squared communication. In *International Conference on Information Security*, pages 314–328. Springer, 2005.
 - [75] Tim Littau and Rihan Hai. Qymera: Simulating quantum circuits using rdbs. In *Companion of the 2025 International Conference on Management of Data*, pages 179–182, 2025.
 - [76] Bin Liu, Fei Gao, Wei Huang, and Qiao-Yan Wen. QKD-based quantum private query without a failure probability. *Science China Physics, Mechanics & Astronomy*, 58(10):100301, 2015.
 - [77] Bingjie Liu, Zhen Qin, Fei Gao, and Qiao-Yan Wen. Decoy-state quantum-key-distribution-based quantum private query. *Physical Review A*, 109:052442, 2024.
 - [78] Minzhao Liu, Ruslan Shaydulin, Pradeep Niroula, Matthew DeCross, Shih-Han Hung, Wen Yu Kon, Enrique Cervero-Martín, Kaushik Chakraborty, Omar Amer, Scott Aaronson, Atithi Acharya, Yuri Alexeev, K. Jordan Berg, Shouvanik Chakrabarti, Florian J. Curchod, Joan M. Dreiling, Neal Erickson, Cameron Foltz, Michael Foss-Feig, David Hayes, Travis S. Humble, Niraj Kumar, Jeffrey Larson, Danylo Lykov, Michael Mills, Steven A. Moses, Brian Neyenhuis, Shaltiel Eloul, Peter Siegfried, James Walker, Charles Lim, and Marco Pistoia. Certified randomness using a trapped-ion quantum processor. *Nature*, pages 1–6, 2025.
 - [79] Marcelo Alejandro Luda, Miguel Antonio Larotonda, Juan Pablo Paz, and Christian Tomás Schmiegelow. Manipulating transverse modes of photons for quantum cryptography. *Physical Review A*, 89(4), April 2014.
 - [80] Anubhav Maitra, Srijita Kundu, Manik Banik, Subhadipa Das, Subhendu Bera, and Indranil Chakrabarty. Device-independent quantum private query. *Physical Review A*, 95:042344, 2017.
 - [81] Sujaya Maiyya, Seif Ibrahim, Caitlin Scarberry, Divyakant Agrawal, Amr El Abbadi, Huijia Lin, Stefano Tessaro, and Victor Zakhary. QuORAM: A quorum-replicated fault tolerant ORAM datastore. In Kevin R. B. Butler and Kurt Thomas, editors, *31st USENIX Security Symposium*, pages 3665–3682, 2022.
 - [82] Frank McKeen, Ilya Alexandrovich, Alex Berenzon, Carlos V. Rozas, Hisham Shafi, Vedvyas Shanbhogue, and Uday R. Savagaonkar. Innovative instructions and software model for isolated execution. In *Proceedings of the 2nd International Workshop on Hardware and Architectural Support for Security and Privacy (HASP)*, 2013.
 - [83] Carlos Aguilar Melchor, Joris Barrier, Laurent Fousse, and Marc-Olivier Killijian. XPIR: Private information retrieval for everyone. *Proceedings on Privacy Enhancing Technologies*, pages 155–174, 2016.
 - [84] Samir Jordan Menon and David J Wu. Spiral: Fast, high-rate single-server pir via the composition. In *2022 IEEE symposium on security and privacy (SP)*, pages 930–947. IEEE, 2022.
 - [85] Pratyush Mishra, Rishabh Poddar, Jerry Chen, Alessandro Chiesa, and Raluca Ada Popa. Oblix: An efficient oblivious search index. In *2018 IEEE Symposium on Security and Privacy (SP)*, pages 279–296. IEEE, 2018.
 - [86] Monetary Authority of Singapore. Quantum key distribution proof-of-concept sandbox for the financial sector (post-implementation report). Technical report, Monetary Authority of Singapore, Singapore, September 2025. Accessed: 2025-11-07.
 - [87] Muhammad Haris Mughees, Hao Chen, and Ling Ren. Onionpir: Response efficient single-server pir. In *Proceedings of the 2021 ACM SIGSAC conference on computer and communications security*, pages 2292–2306, 2021.

- [88] Michael A Nielsen and Isaac L Chuang. *Quantum Computation and Quantum Information*. Cambridge university press, 2010.
- [89] Chao-Yang Pang, Ri-Gui Zhou, Cong-Bao Ding, and Ben-Qiong Hu. Quantum search algorithm for set operation. *Quantum information processing*, 12(1):481–492, 2013.
- [90] Apoorva Patel. Quantum database search can do without sorting. *Physical Review A*, 64(3):34303, 2001.
- [91] Marco Pistoia, Omar Amer, Monik R Behera, Joseph A Dolphin, James F Dynes, Benny John, Paul A Haigh, Yasushi Kawakura, David H Kramer, Jeffrey Lyon, et al. Paving the way toward 800 gbps quantum-secured optical channel deployment in mission-critical environments. *Quantum Science and Technology*, 8(3):035015, 2023.
- [92] Rishabh Poddar, Tobias Boelter, and Raluca Ada Popa. Arx: an encrypted database using semantically secure encryption. *Proc. VLDB Endow.*, 12(11):1664–1678, July 2019.
- [93] Ling Qin and colleagues. Decoy-state quantum private query protocol with two-way communication. *Physica A: Statistical Mechanics and its Applications*, 631:129341, 2024.
- [94] Raghu Ramakrishnan, Johannes Gehrke, and Johannes Gehrke. *Database management systems*, volume 3. McGraw-Hill New York, 2003.
- [95] Mariana F Ramos, Michael Hentschel, Federico Valbusa, Costin Luchian, Martin Achleitner, Alessandro Trenti, Marie-Christine Slater, Mariano Lemus, Thomas Lorünser, and Hannes Hübel. Secure multi-party biometric verification using QKD assisted quantum oblivious transfer. In *ArXiv*, 2501.05327, 2025.
- [96] Xuanle Ren, Le Su, Zhen Gu, Sheng Wang, Feifei Li, Yuan Xie, Song Bian, Chao Li, and Fan Zhang. HEDA: multi-attribute unbounded aggregation over homomorphically encrypted database. *Proc. VLDB Endow.*, 16(4):601–614, 2022.
- [97] David A. Rower, Leon Ding, Helin Zhang, Max Hays, Junyoung An, Patrick M. Harrington, Ilan T. Rosen, Jeffrey M. Gertler, Thomas M. Hazard, Bethany M. Niedzielski, Mollie E. Schwartz, Simon Gustavsson, Kyle Serniak, Jeffrey A. Grover, and William D. Oliver. Suppressing counter-rotating errors for fast single-qubit gates with fluxonium. *PRX Quantum*, 5(4):040342, 2024.
- [98] Sudip Roy, Lucja Kot, and Christoph Koch. Quantum databases. In *CIDR*, 2013.
- [99] Tamer Salman and Yoram Baram. Quantum set intersection and its application to associative memory. *The Journal of Machine Learning Research*, 13(1):3177–3206, 2012.
- [100] Manuel Schönberger, Stefanie Scherzinger, and Wolfgang Mauerer. Ready to leap (by co-design)? join order optimisation on quantum hardware. *Proceedings of the ACM on Management of Data (PACMOD)*, 1(1):1–27, 2023.
- [101] Manuel Schönberger, Immanuel Trummer, and Wolfgang Mauerer. Quantum-inspired digital annealing for join ordering. *Proceedings of the VLDB Endowment (PVLDB)*, 17(3):511–524, 2023.
- [102] Shantanu Sharma, Yin Li, Sharad Mehrotra, Nisha Panwar, Komal Kumari, and Swagnik Roychoudhury. Information-theoretically secure and highly efficient search and row retrieval. *Proceedings of the VLDB Endowment*, 16(10):2391–2403, 2023.
- [103] Si-Jia Sun, Yu-Guang Yang, and Ming-Ou Zhang. Relativistic quantum private database queries. *Quantum Information Processing*, 14(4):1443–1450, 2015.
- [104] Yuanyuan Sun, Sheng Wang, Huorong Li, and Feifei Li. Building enclave-native storage engines for practical encrypted databases. *Proceedings of the VLDB Endowment*, 14(6):1019–1032, 2021.
- [105] Yoshimichi Tanizawa, Akira Murakami, Ririka Takahashi, Kazuaki Doi, Mamiko Kujiraoka, Hideaki Sato, Muneaki Shimada, Nobuo Yaegashi, Shogo Shigeta, Yasunobu Okamura, et al. Quantum key distribution network and quantum secure cloud technologies for genome medicine use-cases. *IEEE Transactions on Quantum Engineering*, 2025.
- [106] Barbara M Terhal and John A Smolin. Single quantum querying of a database. *Physical Review A*, 58(3):1822, 1998.
- [107] Immanuel Trummer. Towards out-of-core simulators for quantum computing. In *Proceedings of the 1st Workshop on Quantum Computing and Quantum-Inspired Technology for Data-Intensive Systems and Applications*, Q-Data ’24, 2024.
- [108] Immanuel Trummer and Christoph Koch. Multiple query optimization on the D-Wave 2X adiabatic quantum computer. *Proceedings of the VLDB Endowment (PVLDB)*, 9(9), 2016.
- [109] I-Ming Tsai, Sy-Yen Kuo, and David S L Wei. Quantum Boolean circuit approach for searching an unordered database. In *Proceedings of the 2nd IEEE Conference on Nanotechnology*, pages 315–318. IEEE, 2002.
- [110] Stephen Tu, M. Frans Kaashoek, Samuel Madden, and Nickolai Zeldovich. Processing analytical queries over encrypted data. *Proceedings of the VLDB Endowment*, 6(5):289–300, 2013.
- [111] Nikolaj Volgushev, Malte Schwarzkopf, Ben Getchell, Mayank Varia, Andrei Lapets, and Azer Bestavros. Conclave: secure multi-party computation on big data. In *Proceedings of the Fourteenth EuroSys Conference 2019*, New York, NY, USA, 2019. Association for Computing Machinery.
- [112] Chao Wang, Wen Yu Kon, Hong Jie Ng, and Charles C-W Lim. Experimental symmetric private information retrieval with measurement-device-independent quantum network. *Light: Science & Applications*, 11(1):268, 2022.
- [113] Chun-Yan Wei, Fei Gao, Qiao-Yan Wen, and Tian-Yin Wang. Practical quantum private query of blocks based on unbalanced-state BB84 quantum-key-distribution protocol. *Scientific Reports*, 4:7537, 2014.
- [114] Robert Wille, Stefan Hillmich, and Lukas Burgholzer. Decision diagrams for quantum computing. In *Design automation of quantum computers*, pages 1–23. Springer, 2022.
- [115] William K Wootters and Brian D Fields. Optimal state-determination by mutually unbiased measurements. *Annals of Physics*, 191(2):363–381, 1989.
- [116] William K Wootters and Wojciech H Zurek. A single quantum cannot be cloned. *Nature*, 299(5886):802–803, 1982.
- [117] Ahmed Younes. Database Manipulation Operations on Quantum Systems. *Quantum Information Review*, 1:9–17, 2013.
- [118] Jian-Liang Zhang, Fu-Zhou Guo, Fei Gao, Bin Liu, and Qiao-Yan Wen. Private database queries based on counterfactual quantum key distribution. *Physical Review A*, 88:022334, 2013.
- [119] Jiaoyi Zhang, Liqiang Peng, Mo Sha, Weiran Liu, Xiang Li, Sheng Wang, Feifei Li, Mingyu Gao, and Huanchen Zhang. Femur: A flexible framework for fast and secure querying from public key-value store. *Proceedings of the ACM on Management of Data*, 3(3):1–29, 2025.
- [120] Shufan Zhang, Xi He, Ashish Kundu, Sharad Mehrotra, and Shantanu Sharma. Secure normal form: Mediation among cross cryptographic leakages in encrypted databases. In *2024 IEEE 40th International Conference on Data Engineering (ICDE)*, pages 5560–5573. IEEE, 2024.
- [121] Zhou Zhang, Song Bian, Zian Zhao, Ran Mao, Haoyi Zhou, Jiafeng Hua, Yier Jin, and Zhenyu Guan. ArcEDB: An arbitrary-precision encrypted database via (amortized) modular homomorphic encryption. In *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security (CCS)*, pages 4613–4627. ACM, 2024.
- [122] Wenting Zheng, Ankur Dave, Jethro G. Beekman, Raluca Ada Popa, Joseph E. Gonzalez, and Ion Stoica. Opaque: An oblivious and encrypted distributed analytics platform. In Aditya Akella and Jon Howell, editors, *14th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, pages 283–298, 2017.